

THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

Homework Set #5 and Mid-Term #2

Team Project

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2018
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Due Date: Wednesday November 07, 2018 in my office by 4:30pm

Number of points.....100
Your homework score.....
Your midterm score.....

100/100
Nice Job!

Note: All problems are worth 100 points. Your final score will be the average value

Solutions to be Used with Homework 5 Questions

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Summary

This report contains the solutions to Homework Set #5/Mid-Term #2 which focuses on the calculation of internal properties of fluids. The first problem deals with the refrigeration of methane using a propane refrigeration cycle. The single stage system is provided and properties such as enthalpy, entropy, and compressibility are calculated with the end goal of determining key performance indicators of the equipment in the system as well as the amount of refrigerant needed. The second problem is a derivation of the residual molar heat capacity of n-pentane at a given temperature and pressure. A framework is provided within the problem which is used as a basis for the derivation. The value of the heat capacity is then calculated and verified against a numerical differentiation of the enthalpy function with respect to temperature.

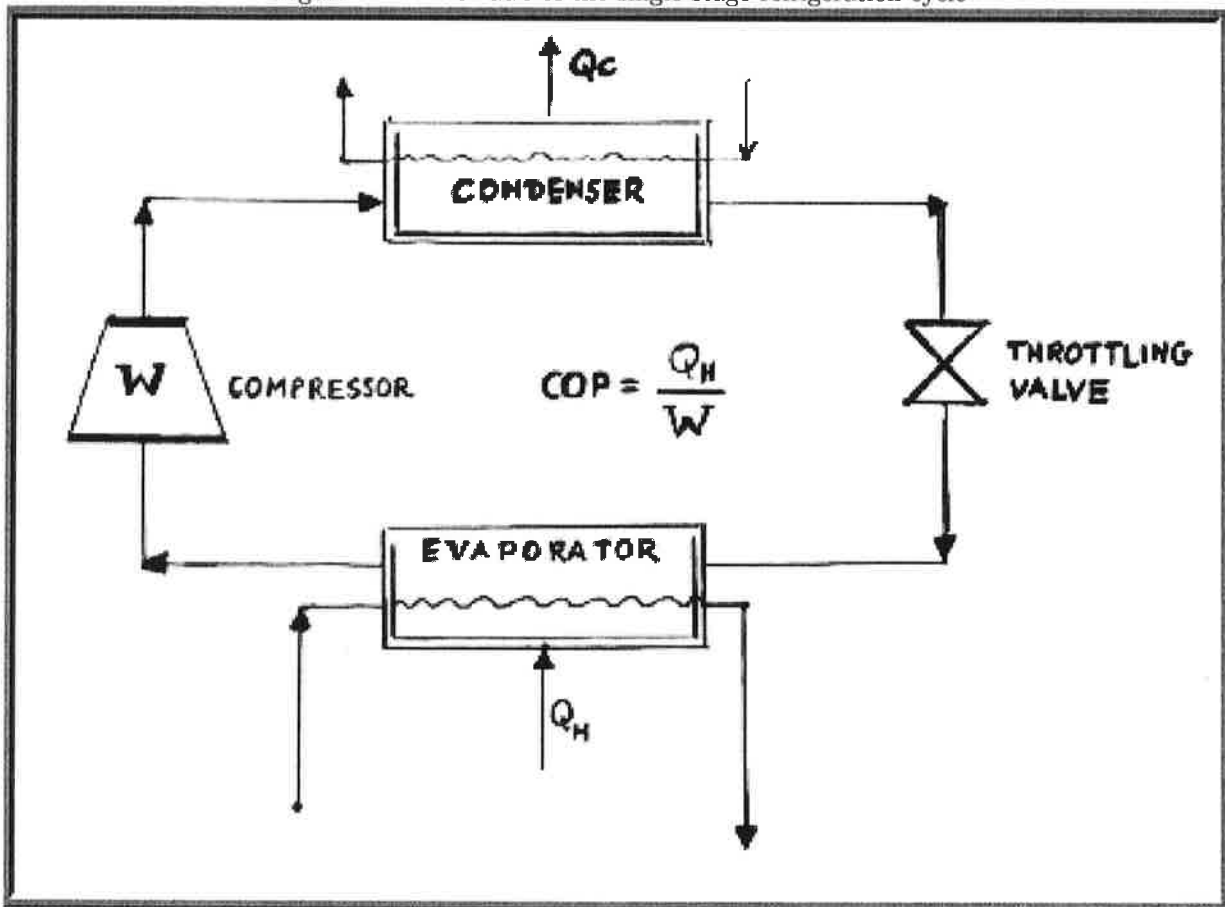
1 Pre-Cooling of Methane in a Liquefaction Proces

The problem objective is to design a single stage refrigeration system using propane that would pre-cool a methane stream from 70 deg Celsius down to a couple of degrees Celsius above the propane saturation temperature. The methane stream conditions are as follows: *Temperature* = 70.0°C, *Pressure* = 27.0 barg, and *Mass Flow Rate* = 70,000 $\frac{kg}{hr}$.

1.1 Design of Single Stage Refrigerator

The refrigerator design consists of a compressor for the propane. The compressed propane goes through a condenser where it is cooled down slightly from the compressor outlet temperature. The propane then goes through a throttling valve to expand it without any work being done on or by the fluid. The final step in the cycle before going back into the compressor is the propane is put through an evaporator where it cools the methane.

Figure 1: A schematic of the single stage refrigeration cycle



1.2 Calculations Involving Equipment and Performance of System

For the refrigeration cycle, certain values are given in order to provide a basis for calculations of pertinent metrics such as enthalpy, entropy, and the coefficients of performance. For uniformity of results it is assumed that at the condenser outlet the propane is saturated liquid at 45°C . It is also assumed that the isentropic efficiency of the compressor is 80% and the evaporator operates at a pressure of 1 bar.

For the calculations, the CHE301 Calculator in Microsoft Excel is used and a table of the results is appended.

At the condenser, the temperature is given as 45°C and that it's a saturated liquid, the EOS_{sat} function of the CHE301 Calculator is used to calculate $P_{sat} = 15.38816 \text{ bar}$. From that, Z_{comp} is calculated and used to calculate the enthalpy of the propane leaving the condenser, $H_{cond} = -117424.06569 \frac{\text{kJ}}{\text{kmol}}$.

At the throttling valve, the gas is expanded isenthalpically to a pressure of 1 bar from the P_{sat} value of 15.38816 bar. The temperature of the throttle outlet is calculated by taking the pressure and using goalseek to match the temperature with the enthalpy of the outlet condenser which is $-117424.06569 \frac{\text{kJ}}{\text{kmol}}$. This method leads to a throttle outlet temperature of $T_{throttle}^{out} = -42.52182^{\circ}\text{C}$.

At the outlet of the evaporator the temperature will still be the same (for propane $T_{throttle}^{out} = -42.52182^{\circ}\text{C}$), and pressure is given to be 1 bar. Therefore, Z_{comp} , enthalpy, and entropy can be calculated at those conditions. For the evaporator outlet, $Z_{comp} = .96691$, $enthalpy = -108548.78506 \frac{\text{kJ}}{\text{kmol}}$, and $entropy = -286.99568 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$.

For the compressor, the enthalpy of the ideal process must be calculated using the ideal condition that change in entropy from the compressor input to output is zero. Given that the pressure of the compressor output is the same as the condenser output, or $P_{sat} = 15.38816 \text{ bar}$, and the entropy of the ideal compressor output is also $-286.99568 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$, the temperature of the compressor output for the ideal situation is calculated in the CHE301 Calculator to be $T_{comp}^{ideal} = 61.22079^{\circ}\text{C}$. This allows for the enthalpy of the ideal compressor output to be $H_{comp}^{ideal} = -102824.34647 \frac{\text{kJ}}{\text{kmol}}$ and, from the enthalpy of the evaporator output, $Work_{ideal} = -102824.34647 - (-108548.78506) = 5724.43859 \frac{\text{kJ}}{\text{kmol}}$.

To extend the results of the ideal compression of propane, the ideal work, $Work_{ideal} = 5724.43859 \frac{\text{kJ}}{\text{kmol}}$ can be divided by the given efficiency $\eta = 0.8$ to calculate $Work_{real} = 7155.54823 \frac{\text{kJ}}{\text{kmol}}$, which is the work put into the compressor to compress the propane. The real work can be used with previous enthalpies and entropies to calculate the real gas temperature at the compressor outlet of $T_{out}^{real} = 76.43079^{\circ}\text{C}$.

Performance Indicator	Value
Power	4146.6685 kJkmol
Duty (Evaporator, Q_H)	5143.26022 kJkmol
Duty (Condenser, Q_C)	-9289.92872 kJkmol

For the above table, the power is calculated by taking the product of the real work and the mass flow rate of propane. The evaporator duty is the product of the change of enthalpy of the propane through the evaporator and the mass flow rate. The condenser duty is the product of the change of enthalpy of propane through the condenser and the mass flow rate of propane.

1.3 Amount of Refrigerant Needed

The amount of refrigerant is dependent on how much cooling needs to be done. For the evaporator, the methane stream going in has a temperature of 70.0°C , flow rate of $70,000 \frac{\text{kg}}{\text{hr}} = 1.212 \frac{\text{kmol}}{\text{s}}$, and a pressure of $27 \text{ barg} = 28.013 \text{ bar}$ given in the problem. From these values Z_{comp} is 0.96441 and $\text{Enthalpy} = -73644.32599 \frac{\text{kJ}}{\text{kmol}}$.

The output methane enthalpy can be calculated by setting the methane temperature to -40°C because this is slightly above the P_{sat} of methane which is approximately -42°C at 28.013 bar . Using the temperature and pressure to calculate the new $Z_{\text{comp}} = 0.86495$, the output methane enthalpy can be calculated to be $-77887.86836 \frac{\text{kJ}}{\text{kmol}}$.

With the input and output methane enthalpies, $\Delta H_{\text{methane}}$ can be calculated and used with the previous $\Delta H_{\text{propane}}$ by the relationship $-\dot{m}\Delta H_{\text{methane}} = \dot{m}\Delta H_{\text{propane}}$, where $\dot{m}$ is $\text{Propane}_{\text{flow}}$. The amount of propane flow to refrigerate the methane is:

$$\text{Propane}_{\text{flow}} = 0.57950 \frac{\text{kmol}}{\text{s}}$$

1.4 Coefficient of Performance for Entire System

The coefficient of performance is given within the diagram as $\text{COP} = \frac{Q_H}{W} = \frac{\text{Energy}_{\text{out}}}{\text{Energy}_{\text{in}}}$ where Q_H , in this case, is equal to the product of the change in enthalpy of the methane stream and its flow rate. W is equal to the product of the work and flow rate for propane. For the refrigeration system:

$$\text{COP} = \frac{-(-4243 \frac{\text{kJ}}{\text{kmol}})(1.2120 \frac{\text{kmol}}{\text{s}})}{(7155.548 \frac{\text{kJ}}{\text{kmol}})(0.5795 \frac{\text{kmol}}{\text{s}})} = 1.24034$$

50/50

2 Calculation of Liquid Heat Capacity under Constant Pressure

The second problem in the homework, n-pentane is given and assumed to follow the van der Waals Equation of State. It is given that the pentane is at a temperature of 25°C and a pressure of 1 bar . The goal of the problem is to verify that n-pentane is liquid at the given conditions; derive the general formula for molar heat capacity at constant pressure; and calculate the molar heat capacity of the pentane given while comparing it to the value obtained through numerical differentiation of the enthalpy function.

2.1 Verification of Liquid State

Using Antoine's equation in the CHE301 Calculator, the saturated vapor pressure of the n-pentane is 0.66625 bar . Since the saturated pressure is lower than the given pressure of 1 bar , the n-pentane is liquid at 1 bar and 25°C .

2.2 Molar Heat Capacity for Van der Waals EOS

For the molar heat capacity at constant pressure, some steps were provided in the problem as checks on the progress of the derivation. The equations are numbered by the step given in the homework problem.

Initially, the difference between constant pressure and constant volume heat capacity is given:

$$C_P - C_V = T \left(\frac{\partial P}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_P \quad (1)$$

From the definitions of the Hemholtz free energy, A , $A = U - TS$ and $dA = -SdT - pdV$ and Maxwell's relations:

$$\left(\frac{\partial V}{\partial T} \right)_P = - \left(\frac{\partial p}{\partial T} \right)_V \left(\frac{\partial V}{\partial P} \right)_T^{-1}$$

This can be directly substituted into equation 1 which gives the equation:

$$C_P - C_V = -T \frac{\left[\left(\frac{\partial P}{\partial T} \right)_V \right]^2}{\left(\frac{\partial P}{\partial V} \right)_T} \quad (2)$$

From there, C_V is needed explicitly. Starting with the definition $C_V = \left(\frac{\partial U}{\partial T} \right)_V$, the derivative with respect to V can be taken for both sides and the order for the energy side can be flipped giving:

$$\left(\frac{\partial C_V}{\partial V} \right)_T = \frac{\partial}{\partial T} \left(\frac{\partial U}{\partial V} \right)_T \bigg|_V$$

When combined with $\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial P}{\partial T} \right)_V - P$:

$$\left(\frac{\partial C_V}{\partial V} \right)_T = \frac{\partial (-P + T(\partial P/\partial T)_V)}{\partial T} \bigg|_V = T \left(\frac{\partial^2 P}{\partial T^2} \right)_V$$

Therefore:

$$\left(\frac{\partial C_V}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_V \quad (3)$$

At constant T , equation 3 becomes:

$$C_V = T \int \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV \quad (4)$$

Using the van der Waals EOS in the form:

$$P = \frac{RT}{V-b} - \frac{a}{V^2}$$

From this equation, $\frac{\partial^2 P}{\partial T^2} = 0$. Therefore equation 4 becomes:

$$\boxed{C_V^R = 0} \quad (5)$$

Using the van der Waals EOS again, $\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V-b}$ and $\left(\frac{\partial P}{\partial V}\right)_T = \frac{2a}{V^3} - \frac{RT}{(V-b)^2}$ and substituting them into equation 2 gives:

$$C_P - C_V = -T \frac{\left(\frac{R}{V-b}\right)^2}{\left(\frac{2a}{V^3} - \frac{RT}{(V-b)^2}\right)}$$

It is known for the van der Waals EOS that $a = \frac{\Lambda(RT)^2}{P}$, $b = \frac{BRT}{P}$, and $V = \frac{ZRT}{P}$. Substituting these into the previous $C_P - C_V$ equation gives:

$$C_P - C_V = -T \frac{\left(\frac{R}{\frac{ZRT}{P} - \frac{BRT}{P}}\right)^2}{\left(\frac{2\left(\frac{\Lambda(RT)^2}{P}\right)}{\left(\frac{ZRT}{P}\right)^3} - \frac{RT}{\left(\frac{ZRT}{P} - \frac{BRT}{P}\right)^2}\right)}$$

This is for real gases. Using $C_P^{IG} - C_V^{IG} = R$ and combining and rearranging terms gives:

$$C_P - C_V = \frac{-RZ^3}{-Z^3 + 2A(Z-B)^2} - R$$

Directly using $C^R = C^R - C^{IG}$ for C_P and C_V and solving for C_P^R :

$$\boxed{C_P^R = R \frac{2A(Z-B)^2}{Z^3 - 2A(Z-B)^2}} \quad (6)$$

2.3 Calculation of Molar Heat Capacity from Given Data

From the problem, n-pentane is given at 1 bar and 25°C. From this and the van der Waals EOS, $A = 0.421875 \frac{P_r}{T_r^2}$ and $B = 0.125 \frac{P_r}{T_r}$. For n-pentane at the given conditions, $P_r = 0.02968$ and $T_r = 0.6349$. Putting these together gives $A = 0.03106$ and $B = 0.00584$ and, from the CHE301 Calculator, $Z = 0.00779$. Plugging these into equation 6 gives:

$$\boxed{C_P^R = \left(8.314 \frac{kJ}{kmol \cdot K}\right) \frac{2(0.03106)(0.00779 - 0.00584)^2}{(0.00779)^3 - 2(0.03106)(0.00779 - 0.00584)^2} = 8.363 \frac{kJ}{kmol \cdot K}}$$

Given $C_P^R = C_P^R - CP^{IG}$ and $C_P^{IG} = \left(\frac{\partial H}{\partial T}\right)_P = \frac{\Delta H}{\Delta T}$ for small ΔT . T_2 can be set to 25.1°C so that ΔT is comparatively small. Enthalpies are then calculated for 25°C and 25.1°C and applied to C_P^{IG} . Therefore:

$$C_P^{IG} = \frac{(-146525.88944) - (-146537.90560)}{.1} = 120.16156 \frac{kJ}{kmol \cdot K}$$

Using this result with the previous C_P^R and $C_P = C_P^R + C_P^{IG}$ gives:

$$C_P = 8.363 + 120.162 = 128.525 \frac{kJ}{kmol \cdot K}$$

This is on the same order of magnitude as the experimental value of $167.19 \frac{kJ}{kmol \cdot K}$

Cp liquid

Appendix A Copy of Original Problem Set

Problem 1: Pre-Cooling of Methane in a Liquefaction Process

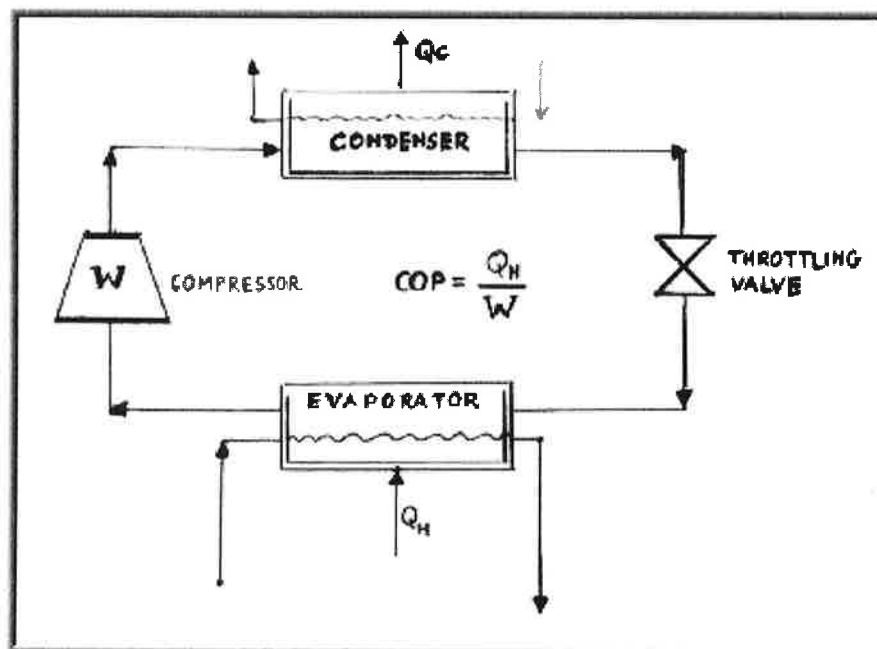
The problem objective is to design a single stage refrigeration system using **propane** that would pre-cool a methane stream from 70 deg Celsius down to a couple of degrees Celsius above the propane saturation temperature. The stream conditions are given in the table below:

Methane Stream Conditions	
Temperature, deg C	70.0
Pressure, barg	27.0
Mass Flow Rate, kg/hr	70,000

1. Design a single stage refrigeration system using **propane** as the refrigerant of choice.
2. List all the major equipment and calculate their performance indicators.
 - a. *Compressor: Performance indicator: Power*
 - b. *Condenser: Performance indicator: Duty*
 - c. *Evaporator: Performance indicator: Duty*
3. Calculate the amount of refrigerant needed
4. Calculate the coefficient of performance for your refrigeration system.

For all your calculations assume that **propane** and **methane** follow the Peng-Robinson Equation of State.

For uniformity of results assume that at the condenser outlet the propane is saturated liquid at 45 deg C. You may also assume that the isentropic efficiency of the compressor is 80%. Finally assume that the evaporator operates at a pressure of 1 bar.



Problem 2: Calculation of Liquid Heat Capacity under Constant Pressure

Assuming that n-pentane follows the van der Waals Equation of State calculate its molar heat capacity under constant pressure at a temperature of 25 deg C and pressure 1 bar. Verify your calculation by performing a numerical differentiation of the enthalpy function with respect to temperature.

1. First verify that at these conditions n-pentane is liquid
2. Second develop a general formula for the molar heat capacity for the van der Waals Equation of State (You may follow the steps outlined below)
3. Finally make the appropriate numerical substitutions in your formula to calculate the molar heat capacity

Follow the steps below(optional) to arrive at the solution:

1. Start with the formula from Chapter 2 page 5a. [No need to prove this formula, but study the derivation to convince yourselves that it is true]

$$C_p - C_v = T \left(\frac{\partial P}{\partial T} \right)_v \left(\frac{\partial V}{\partial T} \right)_p$$

2. Prove:

$$C_p - C_v = -T \frac{\left[\left(\frac{\partial P}{\partial T} \right)_v \right]^2}{\left(\frac{\partial P}{\partial V} \right)_T}$$

3. Prove:

$$\left(\frac{\partial C_v}{\partial V} \right)_T = T \left(\frac{\partial^2 P}{\partial T^2} \right)_v$$

4. At constant T equation in Step 3 becomes

$$dC_v = T \int \left(\frac{\partial^2 P}{\partial T^2} \right)_v dV$$

5. From Step 4 obtain an expression for the residual heat capacity C_p^R
6. Combine all steps to obtain the following formula for the residual heat capacity C_p^R

$$C_p^R = R \frac{2A(Z-B)^2}{Z^3 - 2A(Z-B)^2}$$

Appendix B Table of Values Calculated in Section 1.2

Engineering Calculations											
Molar Composition		Units	Condenser (exit)	Leaving Throttle	Exit Evaporator	Compressor (Exit) Ideal	Compressor (Exit) Real	CH ₄ (in)	CH ₄ (out)		
C ₃ H ₈	mol fraction			Delta H Condenser (Propane)	-9289.92872		Power	4146.66850			
CH ₄											
				Delta H Propane	8875.28063	Duty (Evaporator) Q _H	5143.26022				
				Delta H Methane in the Evaporat	-4243.54237	Duty (Condenser) Q _C	-9289.92872				
				Mass flow rate of Propane	0.57950	Total	9289.92872				
				COP	1.24034						
Conditions and Proj											
		Units									
Molar Flow	kmol/s										
Mass Flow	kg/s									1.21202	
Volume Flow	m ³ /s										
Temperature	C		45.00000	42.52182	42.52182	61.22079	76.43079	70.00000	-40.00000		
Pressure	bar		15.38816	1.00000	1.00000	15.38816	15.38816	28.01300	28.01300		
Vapor Fraction		Z _{comp}	0.05516		0.96691	0.78818	0.81944	0.96441	0.86495		
Specific Enthalpy	kJ/kmol		-117424.06569	-117424.06569	-108548.78506	-102824.34647	-101393.23683	-73644.32599	-77887.86836		
Specific Entropy	kJ/(kmol K)				-286.99568	-286.99568					
Density	kmol/m ³										
Molecular Weight	kg/kmol										
Enthalpy Flow	kJ/s						Ideal Work				
Entropy Flow	kJ/(s K)						Actual Work	0.80000			
Heat	kJ/s						7155.54823				
Power	kW						Efficiency of Compressor				
Efficiency							0.80000				